

Brown Dwarfs to Exoplanets

Brown dwarfs are the lowest mass product of star formation. They are typically very faint compared to stars. Brown dwarfs are similar to exoplanets with comparable temperature and radii, and sometimes mass and age. Brown dwarfs and giant exoplanets have low temperature and complex atmospheres. Exoplanets are more difficult to image than brown dwarfs due to glare from stars. There are currently only approximately 30-40 directly imaged exoplanets. However, there are more than 3000 imaged brown dwarfs. Due to their similarity in characteristics, studying exoplanets allows us to better understand brown dwarfs.



Fig 1: A comparison of brown dwarfs to Jupiter. Brown dwarfs are powerful analogs for giant planets because of their similar physical characteristics. Brown dwarfs fall within the M, L, T and Y spectral classes .

Complex Atmospheres

Once temperatures reach below temperatures below ~2200K, clouds begin to form in the atmospheres of brown dwarfs, and depending on the temperature, the composition of the clouds changes. For example in L dwarfs there are clouds composed of molten iron droplets and silicate clouds, and in T dwarfs there are salt and sulfide clouds. We can study brown dwarfs by monitoring their brightness over time, which is called variability monitoring. Atmospheric features such as clouds, can cause dips in the observed light curve. With this study, we will be able to use these light curves to get information from them we have never been able to before. To date, there have been approximately 70 brown dwarfs monitored for variability (Vos et. al (2020)). Brown dwarfs in the L/T transition are easier to monitor for variability because of rapid rotations and atmospheric inhomogeneities.

Methodology

The aim of this project will be to use light curve mapping techniques on existing light curves of brown dwarfs and directly-imaged exoplanets to reveal the spatial structure of clouds in brown dwarf atmospheres. This will allow us to maximize the use of light curves. We use Starry, a code developed by Luger et al. (2019), which has primarily been used on stellar light curves until now. Starry is a python package which uses light curves as an input to generate surface maps. To do this, we use simulated light curve data from Tan & Showman (2021).The simulated data is atmospheric dynamical simulations that include rotation period, inclination angle, and flux. The simulations produce evolving maps over time as well as light curves for BDs/exoplanets with various properties. Each simulation lasts for 20 rotations.

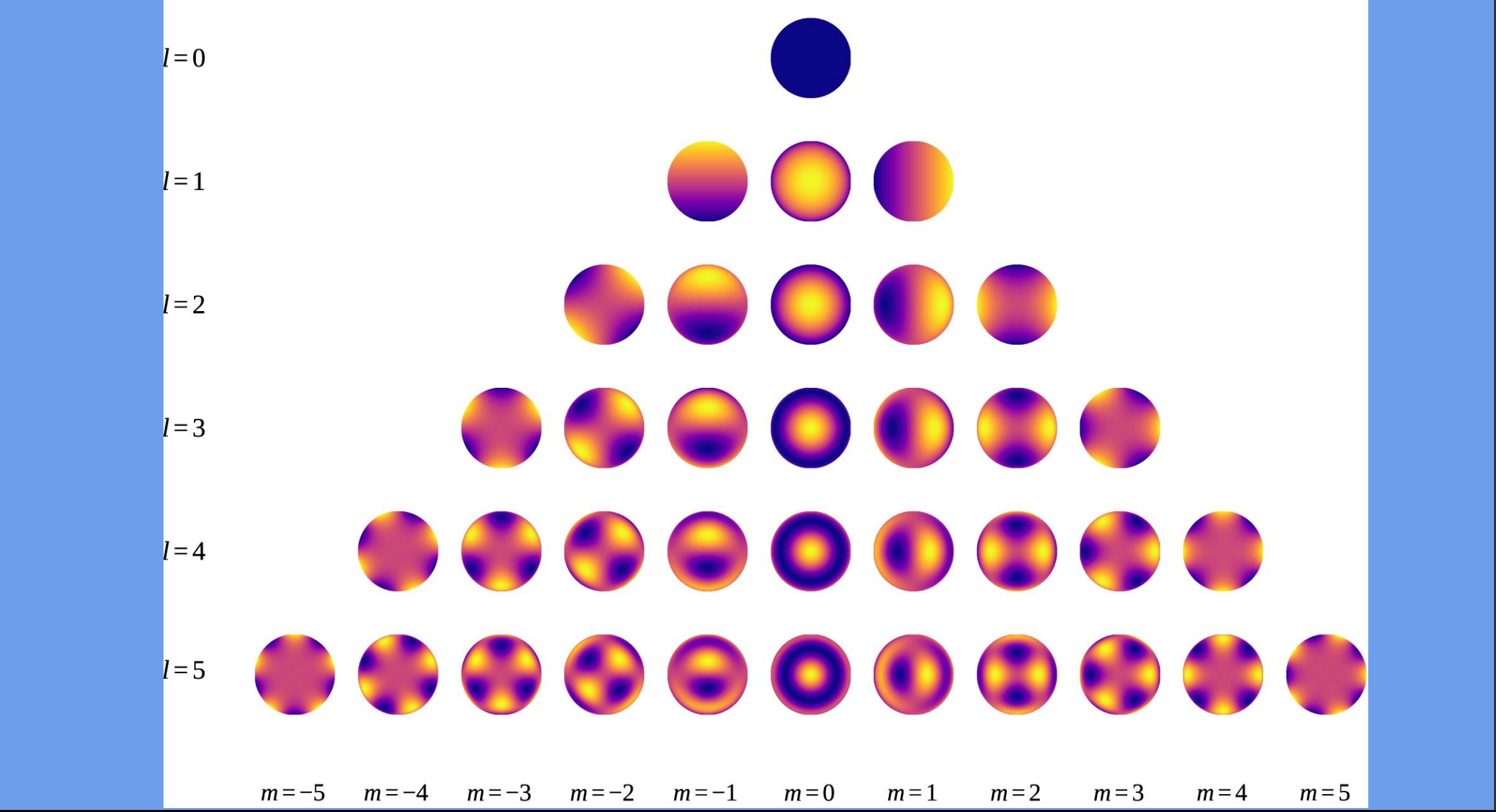


Fig 2: Starry uses spherical harmonics to generate surface maps.

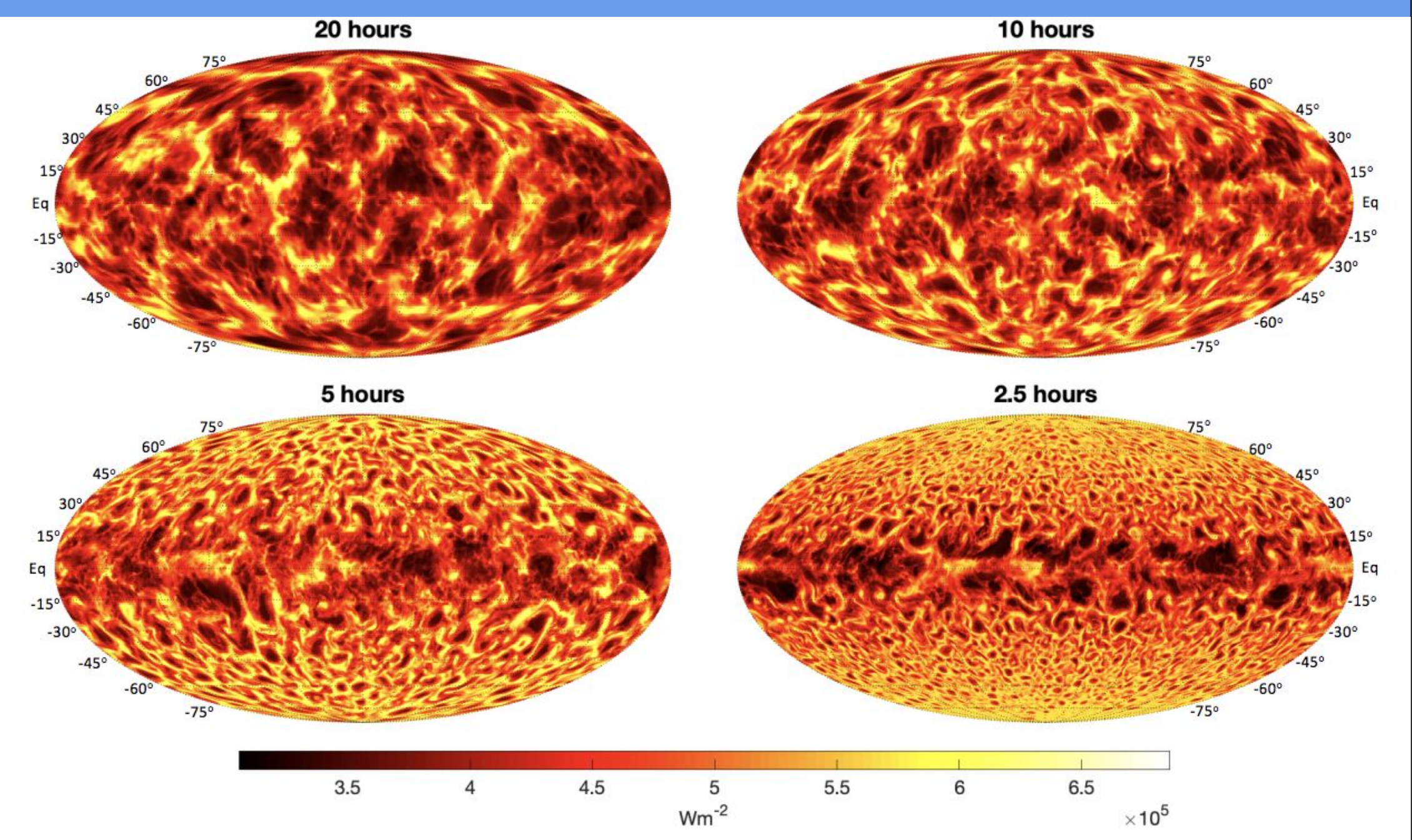


Fig 3: The surface maps generated from the simulation. These maps are our comparison point to test Starry's accuracy.

Results

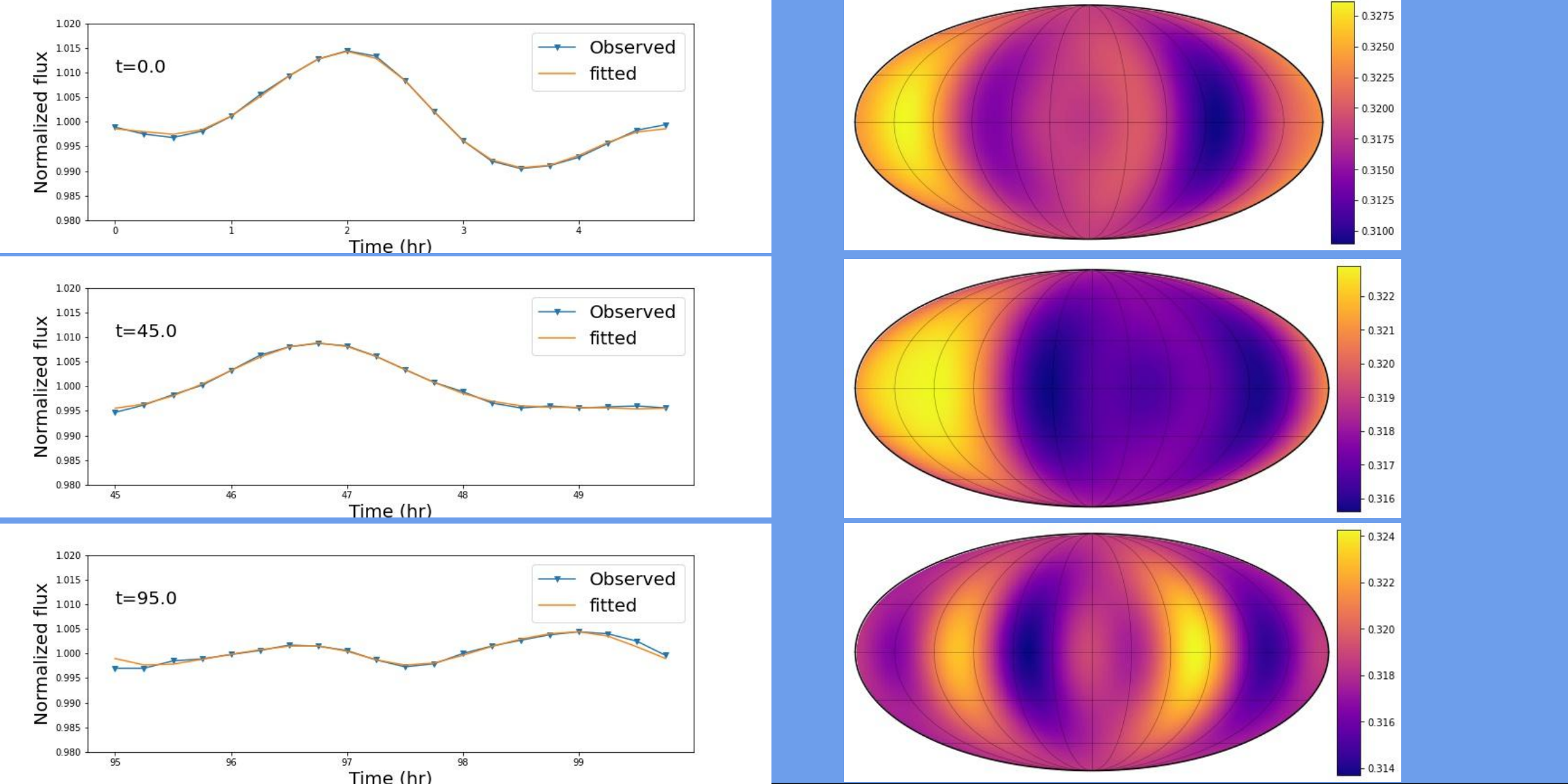


Fig 4: The left panel shows simulated light curves (blue) as well as starry fits (orange) for a brown dwarf with a rotation period of 5 hours at 0 degrees (or equator-on). The right panel shows starry surface maps. They are at 3 different periods of time, starting at 0, 45 and 90 hours. Starry can help us to trace atmospheric features over time. Between this three maps, we can see that the complexity of the surface maps increase over time. It also allows us to examine the intensity values of each point on the map.

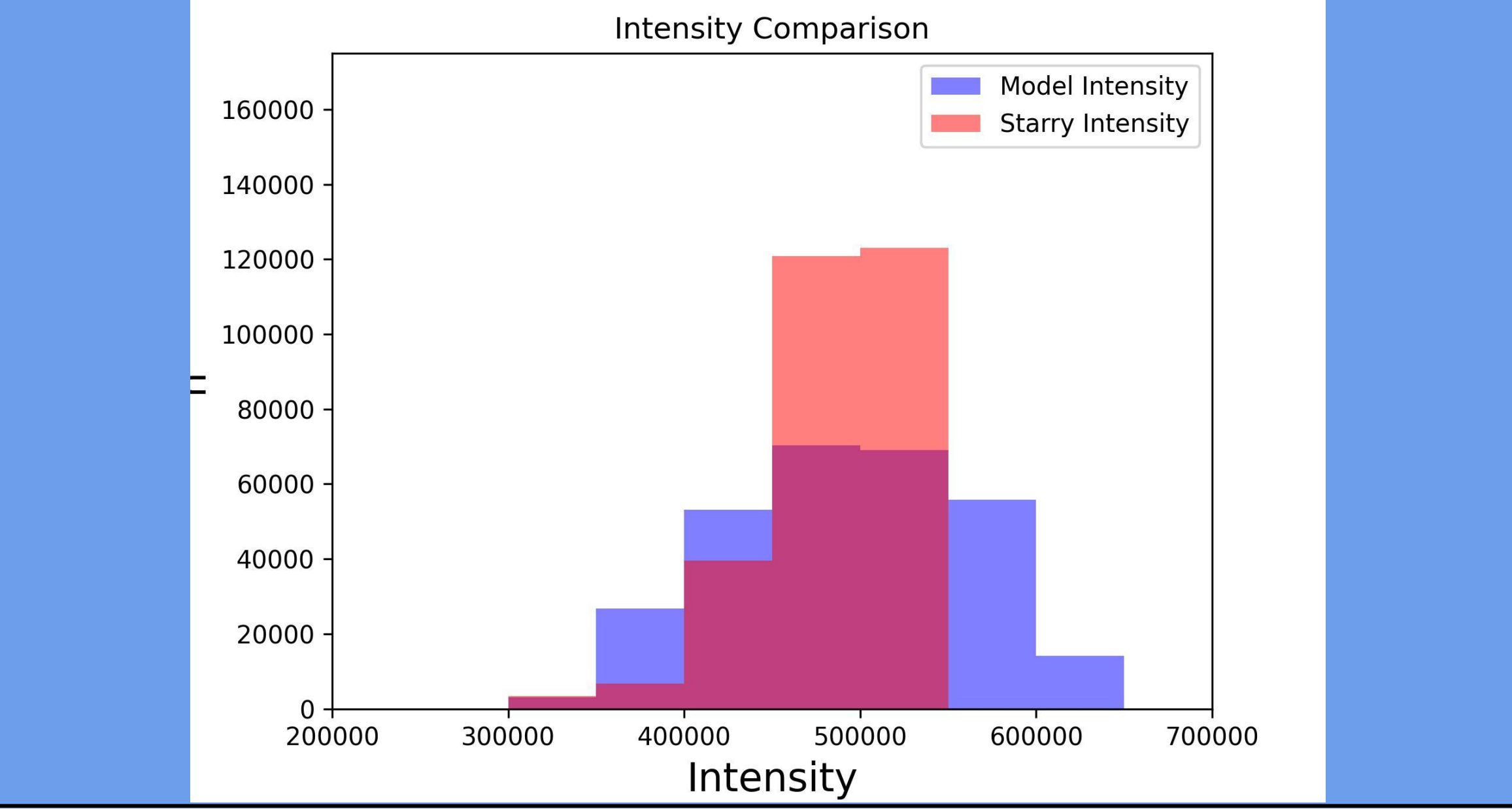


Fig 5: This plot shows a distribution of the intensity values as calculated by Starry. The distribution allows us to compare the intensity values with that of the original map.

Conclusion

From the generated maps, we can see that Starry can create maps that represent the light curves. The next step will be to use Starry's capabilities to try to characterize star spots and clouds.

References

Luger, R., et al. "starry: Analytic Occultation Light Curves." *The Astrophysical Journal* 157.2 (2019): 64.
Vos, J., et al. "Spitzer Variability Properties of Low-gravity L Dwarfs." *The Astronomical Journal* 160.1 (2020): 20.
Tan, X. et al. "Atmospheric circulation of brown dwarfs and directly imaged exoplanets driven by cloud radiative feedback: effects of rotation." *Monthly Notices of the Royal Astronomical Society* 502.1 (2021): 678.
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